

Material Specification Sheet

Name: PEEK, PolyEtherEther-Ketone

Supplier: Plastics Supplier Company

Material description

PEEK is a high-performance engineering thermoplastic. It is strong, rigid, and highly resistant to chemicals, water, steam, and elevated temperatures. Because of this combination of properties, PEEK is commonly used in demanding environments where ordinary plastics would not perform well. In some harsh-service applications, it may also be used as a strong alternative to fluoropolymers.

Main benefits

PEEK provides strong chemical resistance, low moisture uptake, good wear and abrasion resistance, high mechanical strength, and reliable performance in hot water or steam. It also has a UL94 V-0 flammability rating.

Common applications

Typical application areas include automotive, marine, electronics, medical, aerospace, oil and gas, and alternative energy.

Key properties

Property	Value
Density	~1.31-1.32 g/cm ³
Water absorption, 24 h	0.10-0.15%
Tensile strength	14,500-16,000 psi, unfilled PEEK
Tensile modulus	450,000-550,000 psi, unfilled PEEK
Flexural strength	24,000-25,000 psi, unfilled PEEK
Compressive strength	17,000-20,000 psi, unfilled PEEK
Melting temperature	644°F / 340°C
Maximum operating temperature	480°F / 249°C, unfilled PEEK
Volume resistivity	4.9 x 10 ¹⁶ ohm-cm, extruded unfilled PEEK